

29 Ans: C

This is actually an E-field question. PE is assumed to be zero initially.

$$\begin{array}{ll}\text{For Particle 1,} & \text{Initial KE}_1 + \text{Initial PE} = \text{Final KE} + \text{Final PE} \\ & 4.0 \text{ MeV} + 0 = 0 + Qq / 4\pi\epsilon_0 d\end{array}$$

$$\begin{array}{ll}\text{For Particle 2,} & \text{Initial KE}_2 + \text{Initial PE} = \text{Final KE} + \text{Final PE} \\ & \text{Initial KE}_2 + 0 = 0 + Qq / 4\pi\epsilon_0(2d) \\ & \text{Initial KE}_2 = 4.0 / 2 = 2.0 \text{ MeV}\end{array}$$

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